Statistical Analysis Report: Income and Sex in the 1994 U.S. Census

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Dataset: UCI Adult (Census Income) training extract (32,561 rows, 15 columns)

Project: Conduct a Statistical Analysis Using Python capstone (Adult income analysis)

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Overview

I analyzed the UCI Adult (Census Income) dataset (Becker & Kohavi, 1996) to test whether sex is associated with earning over $50K. The dataset is publicly available at https://archive.ics.uci.edu/dataset/2/adult. 7,841 of its 32,561 records (24.08%) earn over $50K.

This report covers the descriptive statistics, four visual models, and one chi-square test of independence. The end result is a clear difference between men and women, with men earning $50K far more often than women, 30.57% against 10.95%. According to the chi-square test this gap is far too large to be sampling noise alone, with a p-value of about 1e-332. The effect size, Cramér's V, is 0.2160, a small-to-moderate association.

Dataset Description

As noted above in the overview, the analysis uses the UCI Adult training extract (Becker & Kohavi, 1996). This dataset is a real public extract of the 1994 U.S. Census, hosted by the UCI Machine Learning Repository, in it each row is one person, with the income column recording whether that person earns over $50K per year.

Dataset structure

The table contains 32,561 rows and 15 columns, with the following columns and columns type:

Column type	Count	Columns
Numeric	6	age, fnlwgt, education-num, capital-gain, capital-loss, hours-per-week
Categorical	9	workclass, education, marital-status, occupation, relationship, race, sex, native-country, income

The test uses the two categorical columns, sex and income. The descriptive statistics follow age and hours-per-week, the two numeric columns most relevant to the outcome.

Missing and duplicate values

Three columns carry missing values (workclass 1,836, occupation 1,843, and native-country 583, 4,262 cells in total). The sex and income columns are complete: all 32,561 rows carry a recorded value for both. The test below therefore uses every row. The missing cells in the other columns do not affect it.

The load check also flags 24 exact duplicate rows. I keep them, because a census extract can legitimately hold two people with identical recorded attributes.

Methods

The initial screening

Lusa et al. (2024) describes initial data analysis as a step before inference, including reviewing missing values and examining variable distributions. Although this paper discussion focuses on longitudinal studies, it's checks can help characterize datasets before testing.

The screening for this analysis covered:

- Column types, missing values, and exact duplicate rows.
- Descriptive statistics for age and hours-per-week.

The hypothesis test

A chi-square test of independence was used because both sex and income are categorical. Here chi-square approximation is appropriate when expected cell counts are sufficiently large (Agresti, 2007); it's smallest expected count in this table is reported in the Results.

The contingency table is unweighted so each record contributes one count. It's inferential results should not be treated as design-adjusted estimates for the U.S. population as the analysis does not adjust for the dataset's sampling weights or a complex survey design.

The hypotheses and significance level are:

- Null hypothesis (H0): sex and income are independent.
- Alternative hypothesis (H1): sex and income are associated.

- Significance level: alpha = 0.05, set before the test.

Direction and assumptions

In order to describe the direction of the association I used the observed counts and over $50K rates. The reported 95% confidence interval for the difference in over $50K rates is a normal-approximation interval for the difference in sample proportions, calculated as male minus female.

Independent observations

The relationship categories suggest that some records may come from people in the same family or household, but the available data do not identify household groupings. I therefore cannot verify independence or adjust the analysis for possible clustering. The reported p-value and confidence interval are unadjusted for this possibility and should be interpreted with caution (Agresti, 2007). The Limitations section discusses this further.

The correction and the effect size

For a two-by-two table, SciPy's chi2_contingency applies the Yates continuity correction by default. Results give SciPy's Yates-corrected statistic for comparison. The uncorrected Pearson chi-square statistic and its p-value was also reported.

Cramér's V was calculated from the uncorrected Pearson statistic in order to describe the association strength. The p-value measures neither that strength nor the probability that the null hypothesis is true. V squared is not interpreted as the percentage of income variation explained by sex.

Results

The descriptive statistics show the shape and balance of the four relevant variables:

- age is right-skewed:
    - mean is 38.5816
    - median is 37.0
    - standard deviation is 13.6404

- interquartile range is 28.0 to 48.0
- skewness is 0.5587

- hours-per-week concentrates at a 40-hour week:
- mean is 40.4375
- median is 40.0
- standard deviation is 12.3474
- interquartile range is 40.0 to 45.0
- skewness is 0.2276

- income is unbalanced, with 7,841 people (24.08%) over $50K.

- sex is uneven, with 21,790 men and 10,771 women.

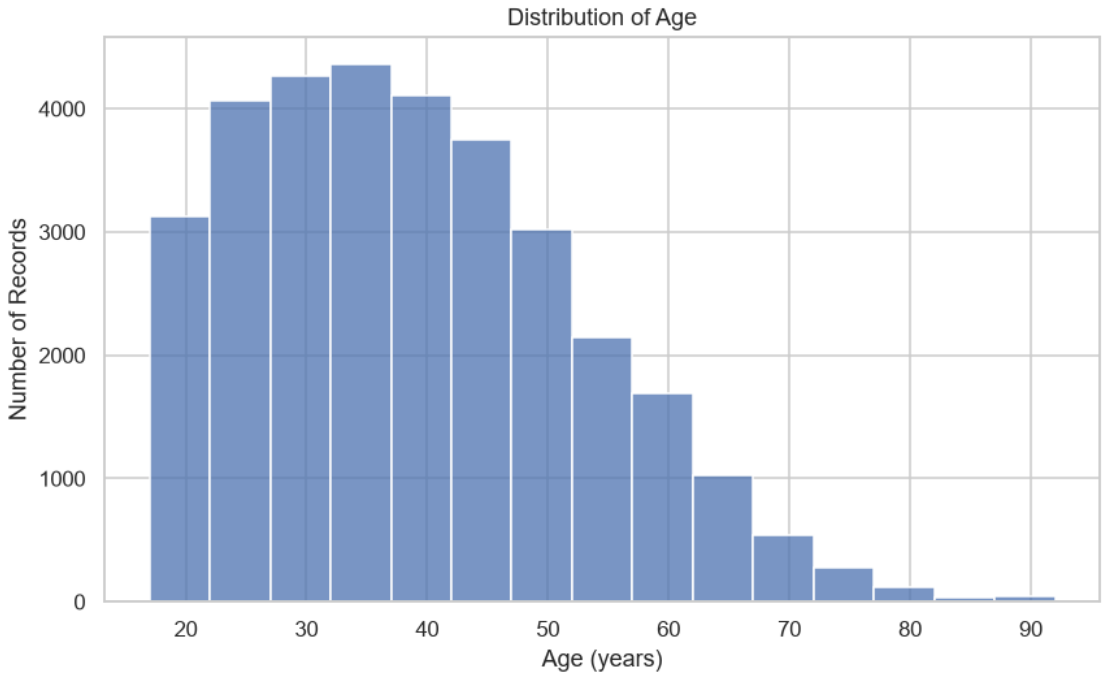


Figure 1. Distribution of age, showing the right skew.

The notebook's supporting counts show it holds 47 records, with 43 at the 90 cap, 3 at 88, and 1 at 87. The data appear top-coded at 90 (respondents beyond the cap recorded as 90), so

that bump reflects the cap rather than a natural peak of older respondents.

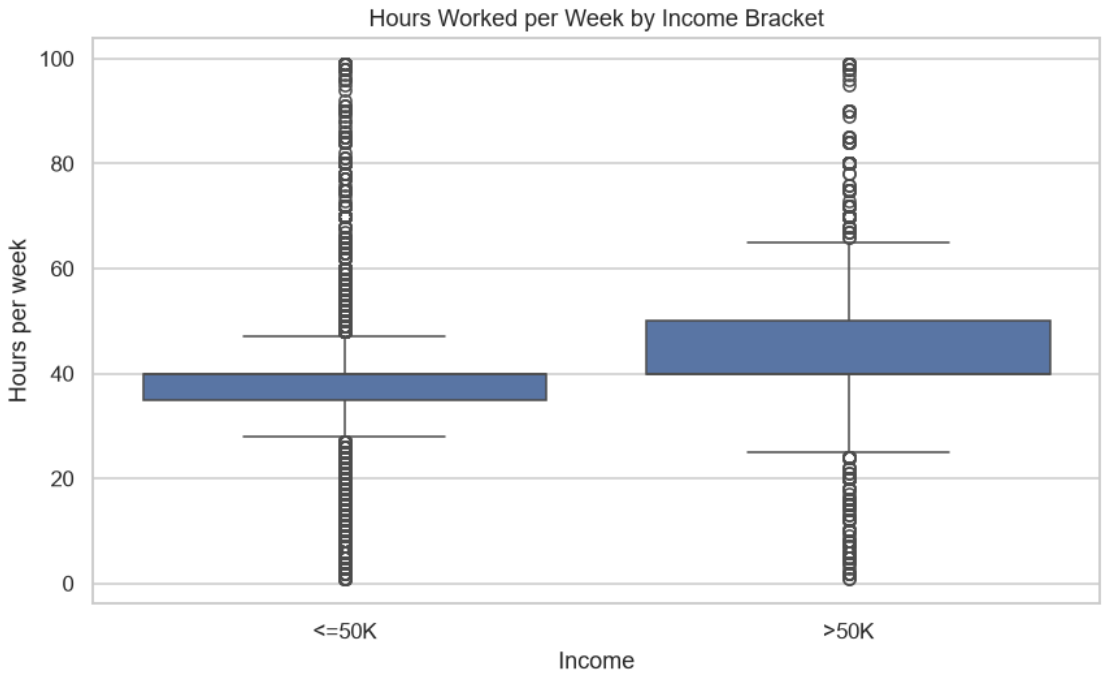


Figure 2. Hours worked per week by income bracket.

- **Median:** 40 hours in both groups.
- **Middle range:** 40 to 50 hours in the over-$50K group, compared with 35 to 40 hours in the at-or-below-$50K group.
- **Past 40 hours:** 49.2% in the over-$50K group, compared with 23.2% in the at-or-below-$50K group.

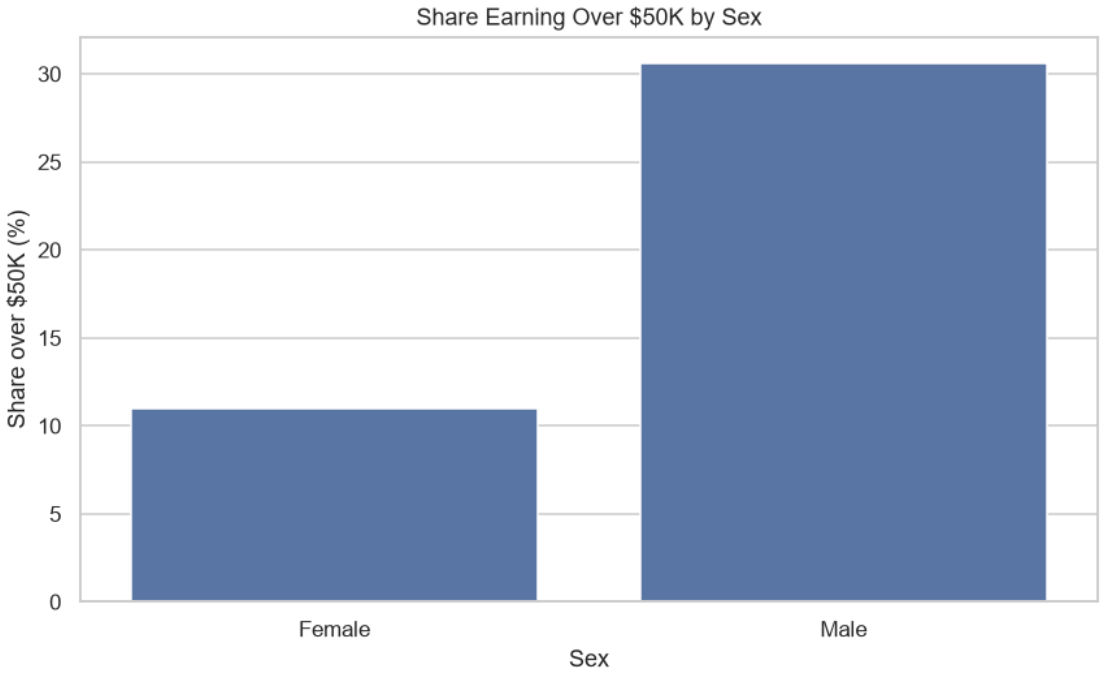


Figure 3. Share earning over $50K by sex.

Figure 3 shows the income rate by sex, the clearest single view of the question. 30.57% of men earn over $50K against 10.95% of women, a gap of 19.63 percentage points.

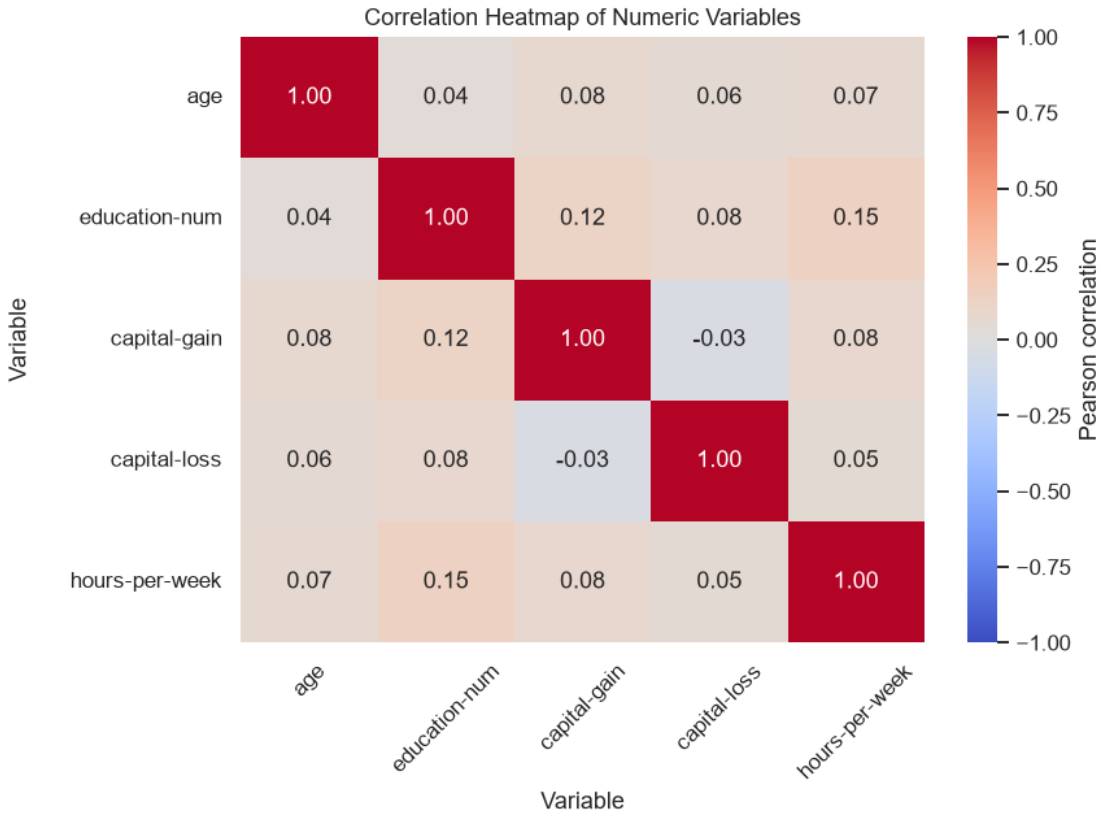


Figure 4. Correlation heatmap of the numeric variables.

- education-num and hours-per-week are the strongest link: 0.1481.
- income is categorical and outside the matrix, so this view does not directly answer the income question.
- fnlwgt is omitted because it is a census sampling weight, not a measured attribute.
- capital-gain and capital-loss are mostly zeros (91.67% and 95.33%), so their correlations are not informative.

The chi-square test rejects independence.

- Pearson statistic: 1518.8868; Yates-corrected: 1517.8134; degrees of freedom: 1.
- Uncorrected Pearson p-value: about 1e-332, far below 1e-300.
- Smallest expected count: 2593.76, meeting the expected-count condition for the chi-square

approximation.

- Effect size: Cramér's V 0.2160; V squared 0.0466.

The contingency table is:

Sex	At or below $50K	Over $50K
Female	9,592	1,179
Male	15,128	6,662

The over-$50K rate is 19.63 percentage points higher for men than for women. Its 95% confidence interval spans [18.78, 20.48] (Agresti, 2007). The interval and the p-value both assume independent observations, so possible household clustering makes both approximate.

Interpretation for a Non-Technical Audience

In this dataset, men were much more likely than women to earn over $50K. About three in ten men were above the threshold, against about one in ten women, a gap of about 19.63 percentage points. Sex is associated with the difference, but it does not account for every difference between the groups.

The result is an association, not proof that sex caused the difference in earnings, and it applies to the 1994 United States data. The Limitations section lists the other differences between the groups and the other caveats.

Limitations and Potential Bias

This is an observational comparison, so the result is an association, not proof of sex causing the difference.

The 1994 U.S. Census extract and its undocumented sampling design limit how far the result generalizes. The $50K threshold is nominal and not inflation-adjusted, and sex is recorded as a binary field.

Missing values in three columns and the possibility of household-linked rows make the

observed rates, p-value, and confidence interval approximate. A follow-up model using education-num, occupation, and hours-per-week, accounting for those issues, would be necessary to further analyze and get a more accurate analysis.

References

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